Forecasting Personal Consumption Expenditures on Energy Goods and Services in the US (2010–2025) Using ARIMA Models

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I. Introduction

This study focuses on monthly Personal Consumption Expenditures (PCE) on Energy Goods and Services in the US from January 2010 to September 2025. The objective is to model the dynamics of this series using the ARIMA methodology and to forecast the upcoming periods.

II. Data Description

1) **Source of Data:** The data are obtained from the Federal Reserve Economic Data (FRED) database, seasonally adjusted, monthly from Jan 2010 to Sep 2025. The values are reported in billions of US dollars.
2) **Length of Series:** The dataset contains 189 observations.
3) **Summary Statistics:** Table 1 indicates that the monthly expenditures have a mean of 633.87 billion USD and a standard deviation of 98.32 billion USD, ranging from 415.8 to 914.9 billion USD.

TABLE I
Summary Statistics of Monthly US Energy PCE (2010–2025)

Variable	Obs	Mean	Std. Dev.	Min	Max
energy (Billions of USD)	189	633.87	98.32	415.8	914.9

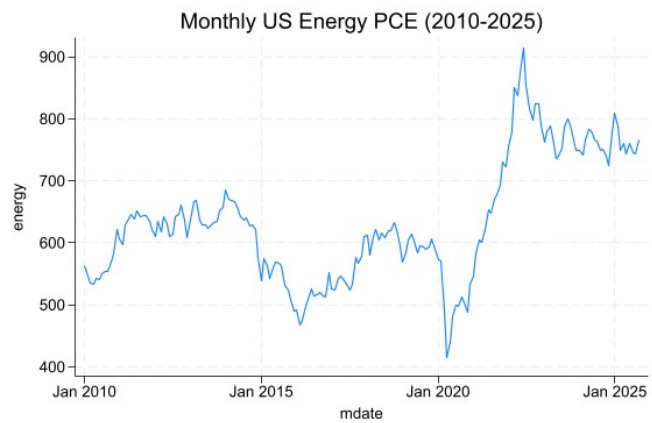

Fig. 1. Monthly US Energy PCE (2010-2025)

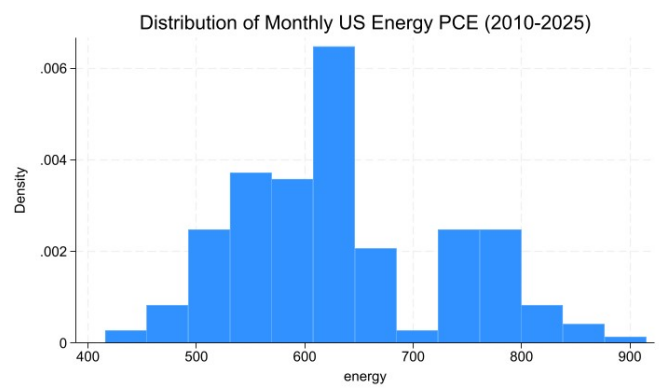

Fig. 2. Distribution of Monthly US Energy PCE (2010-2025)

The line plot (Fig. 1) illustrates the monthly energy consumption expenditures from 2010 to 2025. The series generally shows moderate fluctuations: higher energy expenditures occur in winter and summer months, while lower values appear in spring and autumn. The series does not follow a perfectly linear trend. There is a sharp decline observed in 2020 and a noticeable upward trend after 2020, reflecting the recovery in energy consumption following the COVID-19 pandemic.
The histogram (Fig. 2) shows the distribution of monthly energy expenditures. Most observations are concentrated around 600–700 billion USD, indicating that the series is roughly centered near the mean, with a few extreme values at both ends corresponding to unusually low or high expenditure months.
The ACF plot (Fig. 3) shows a gradual decline of autocorrelations, suggesting a non-seasonal ARIMA process may be appropriate. The PACF (Fig. 4) shows significant

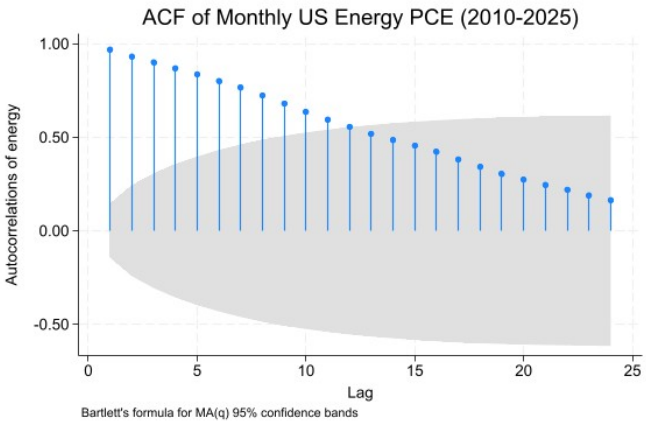

Fig. 3. ACF of Monthly US Energy PCE (2010-2025)

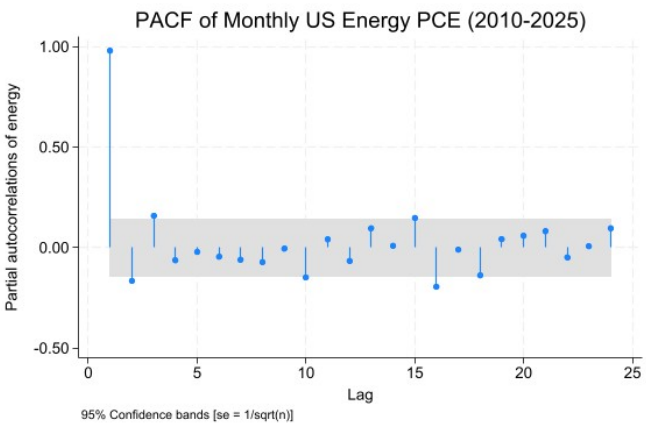

Fig. 4. PACF of Monthly US Energy PCE (2010-2025)

spikes at lag 1, indicating possible short-term autoregressive effects.

III. Analysis

1) **Stationarity (ADF, PP and KPSS test):** To assess the stationarity of the monthly energy consumption series, we performed three complementary tests: the Augmented Dickey–Fuller (ADF) test (Table II), the Phillips–Perron (PP) test (Table III), and the KPSS test (Table IV).
The ADF test with a trend term yielded a test statistic of -1.775, which is well above the 5% critical value of -3.438, with a p-value of 0.7167. Similarly, the PP test produced a Z(t) statistic of -1.899 (p-value = 0.6553), also failing to reject the null hypothesis of a unit root. Both tests therefore suggest that the time series is non-stationary. In contrast, the KPSS test, which has the null hypothesis of stationarity, gave a test statistic of 2.47 at lag 0, exceeding the 1% critical value of 0.216, strongly rejecting the null. This further confirms that the series exhibits a unit root and is non-stationary. Taken together, the three tests consistently indicate that the monthly energy expenditure series is non-stationary, and therefore differencing is required before fitting an ARIMA model.

TABLE II
Augmented Dickey-Fuller (ADF) Test

Observations	188			
ADF Z(t) Statistic	-1.775			
Critical Values	1%: -4.011, 5%: -3.438, 10%: -3.138			
MacKinnon approximate p-value for Z(t) = 0.7167				
Variable	Coefficient	Std. Err.	t	P>|t|
energy L1.	-.0339843	.0191463	-1.77	0.078
_trend	.042779	.0345223	1.24	0.217
_constant	18.55181	10.86918	1.71	0.090

TABLE III
Phillips-Perron (PP) Test

Observations	188			
PP Z(t) Statistic	-1.899			
Critical Values	1%: -4.011, 5%: -3.438, 10%: -3.138			
MacKinnon approximate p-value for Z(t) = 0.6553.				
Variable	Coefficient	Std. Err.	t	P>|t|
energy L1.	.9660157	.0191463	50.45	0.000
_trend	.042779	.0345223	1.24	0.217
_constant	18.55181	10.86918	1.71	0.090

TABLE IV
KPSS Test

Critical Values	10%: 0.119, 5%: 0.146, 2.5%: 0.176, 1%: 0.216
Lag Order	Test Statistic
0	2.47
1	1.26
2	.852
3	.648
4	.526
5	.445
6	.386
7	.343
8	.309
9	.283
10	.261
11	.243
12	.229
13	.216
14	.205

2) **Differenced series:** After applying first differencing to the monthly energy consumption series, we reassessed stationarity using the Augmented Dickey–Fuller (ADF) test (Table V), Phillips–Perron (PP) test (Table VI), and KPSS test (Table VII).
The ADF test with a trend term yields a test statistic of -11.636 (p-value < 0.001), well below the 1% critical value of -4.011. Similarly, the PP test produces a Z(t) statistic of -11.522 (p-value < 0.001), both strongly rejecting the null hypothesis of a unit root.

In contrast, the KPSS test, which has the null of stationarity, gives test statistics well below the 1% critical value (e.g., 0.0654 at lag 0), failing to reject the null. Taken together, these results confirm that the differenced series is stationary, satisfying the requirements for fitting an ARIMA model. This justifies the use of d=1 in the ARIMA specification.

TABLE V
AUGMENTED DICKEY-FULLER (ADF) TEST

Observations	187			
ADF Z(t) Statistic	-11.636			
Critical Values	1%: -4.011, 5%: -3.438, 10%: -3.138			
MacKinnon approximate p-value for Z(t) = 0.0000				
Variable	Coefficient	Std. Err.	t	P>|t|
energy LD	-.8491719	.0729795	-11.64	0.000
_trend	.0071872	.0294633	0.24	0.808
_constant	.3227983	3.193357	0.10	0.920

TABLE VI
PHILLIPS-PERRON (PP) TEST

Observations	187			
PP Z(t) Statistic	-11.522			
Critical Values	1%: -4.011, 5%: -3.438, 10%: -3.138			
MacKinnon approximate p-value for Z(t) = 0.0000				
Variable	Coefficient	Std. Err.	t	P>|t|
energy LD	.1508281	.0729795	2.07	0.040
_trend	.0071872	.0294633	0.24	0.808
_constant	.3227983	3.193357	0.10	0.920

TABLE VII
KPSS TEST

Critical Values	10%: 0.119, 5%: 0.146, 2.5%: 0.176, 1%: 0.216
Lag Order	Test Statistic
0	.0654
1	.0569
2	.0593
3	.0606
4	.0606
5	.0600
6	.0589
7	.0575
8	.0566
9	.0550
10	.0538
11	.0529
12	.0527
13	.0529
14	.0537

IV. MODEL SELECTION (ARIMA)

The autocorrelation function (ACF) and partial autocorrelation function (PACF) of the first-differenced energy series (D.energy) are plotted to guide the selection of AR and MA orders for the ARIMA model. In the ACF plot (Fig. 5), most autocorrelations lie within the 95% confidence bands,

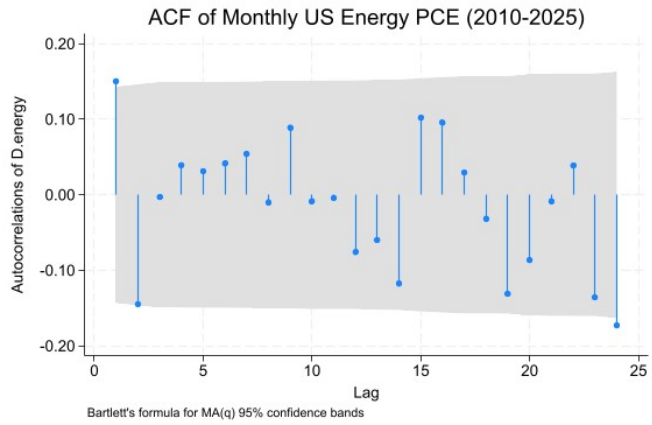

Fig. 5. ACF of Monthly US Energy PCE (2010-2025)

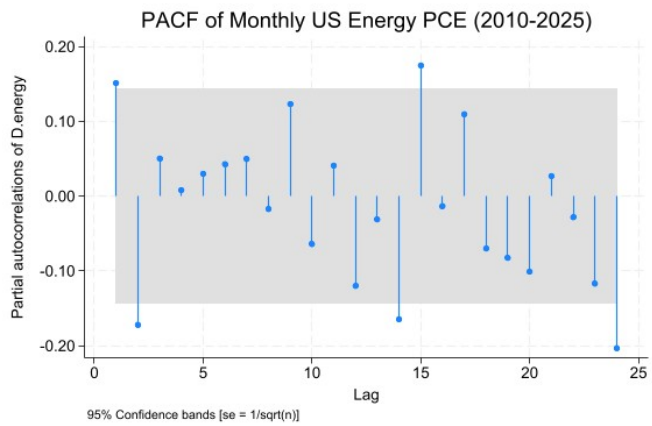

Fig. 6. PACF of Monthly US Energy PCE (2010-2025)

with only one lag slightly exceeding the threshold, indicating limited residual autocorrelation. The PACF plot (Fig. 6) shows a similar pattern, with two lags slightly outside the 95% bands, suggesting minor partial autocorrelation at these points. Based on these diagnostics, two candidate ARIMA models are considered for further evaluation: ARIMA(1,1,1) and ARIMA(2,1,1). Both models will be compared using log-likelihood, AIC/BIC, and residual diagnostics to determine the best specification for forecasting.

1) **ARIMA(1,1,1) Model Estimation**
The ARIMA(1,1,1) model (Table VIII) was estimated on the monthly energy consumption series without a constant term because the stationarity tests showed that the intercept (_cons) was not statistically significant. The estimated parameters are as follows:
AR(1) coefficient: -0.748 ($p < 0.001$),
MA(1) coefficient: 0.924 ($p < 0.001$),
Estimated standard deviation of residuals = 21.08.
All coefficients are statistically significant at the 1% level, indicating strong short-term autocorrelation and moving average effects in the differenced series. The

TABLE VIII
ARIMA(1,1,1) MODEL

Observations		188		
Log likelihood		-840.0578		
Wald $\chi^2(2)$		365.34		
Prob $> \chi^2$		0.0000		
	Coefficient	Std. Err.	z	P> $\lvert z \rvert$
ARMA				
AR L1	-0.7476289	0.0819849	-9.12	0.000
MA L1	0.9236295	0.0549817	16.80	0.000
sigma	21.08247	0.9045066	23.31	0.000

TABLE IX
AKAIKE'S INFORMATION CRITERION AND BAYESIAN INFORMATION CRITERION FOR ARIMA(1,1,1)

Observations	188
Log likelihood	-840.0578
Degrees of freedom	3
AIC	1686.116
BIC	1695.825

model has a log-likelihood of -840.0578, an AIC of 1686.116, and a BIC of 1695.825 (Table IX). These information criteria will be compared with alternative specifications to select the best model.

2) **ARIMA(2,1,1) Model Estimation**
The ARIMA(2,1,1) model (Table X) was estimated on the monthly energy consumption series without a constant for the same reason as in the ARIMA(1,1,1) model. The estimated parameters are as follows:
AR(1) coefficient: -0.081 (p = 0.817, not significant),
AR(2) coefficient: -0.130 (p = 0.180, not significant),
MA(1) coefficient: 0.267 (p = 0.455, not significant),
Estimated standard deviation of residuals = 21.22.
All AR and MA coefficients are statistically insignificant at conventional levels, suggesting that the additional AR term does not improve model fit. The model has a log-likelihood of -841.1112, which is lower than the log-likelihood of -840.057 of ARIMA(1,1,1) model; an AIC of 1690.222, and a BIC of 1703.168 (Table XI), which are higher than the ARIMA(1,1,1) model (AIC = 1686.116, BIC = 1695.825). Model comparison relies on the log-likelihood and information criteria: a higher (less negative) log-likelihood indicates a better fit to the data, whereas lower values of AIC and BIC are preferred as they penalize model complexity. Therefore, ARIMA(2,1,1) does not outperform ARIMA(1,1,1) for this series.
Overall, the results indicate that ARIMA(1,1,1) remains the preferred specification for forecasting monthly energy consumption

V. MODEL DIAGNOSTICS

First, the Portmanteau (Q) test for white noise (Table XII) was applied to the residuals. The test statistic is Q(40) = 47.8355 with a p-value of 0.1846, which means that the null hypothesis of no autocorrelation in the residuals cannot be rejected. This indicates that the residuals behave like white noise and that the model has successfully captured the serial dependence in the data.

Second, the inverse roots of the AR and MA characteristic polynomials (Fig. 7) were examined using the eigenvalue stability condition. All eigenvalues for both the AR and MA components lie strictly inside the unit circle, satisfying the conditions for stationarity (AR part) and invertibility (MA part).

Taken together, these results confirm that the ARIMA(1,1,1) model is well-specified, stable, and suitable for forecasting.

TABLE X
ARIMA(2,1,1) MODEL

Observations		188		
Log likelihood		-841.1112		
Wald $\chi^2(3)$		19.98		
Prob $> \chi^2$		0.0002		
	Coefficient	Std. Err.	z	P> $\lvert z \rvert$
ARMA				
AR L1	-0.080885	0.3498488	-0.23	0.817
AR L2	-0.1297943	0.0967085	-1.34	0.180
MA L1	0.2673433	0.3576337	0.75	0.455
sigma	21.21702	0.9202762	23.06	0.000

TABLE XI
AKAIKE'S INFORMATION CRITERION AND BAYESIAN INFORMATION CRITERION FOR ARIMA(2,1,1)

Observations	188
Log likelihood	-841.1112
Degrees of freedom	4
AIC	1690.222
BIC	1703.168

TABLE XII
PORTMANTEAU TEST FOR WHITE NOISE

Portmanteau (Q) statistic	47.8355
Prob $> \chi^2$ (40)	0.1846

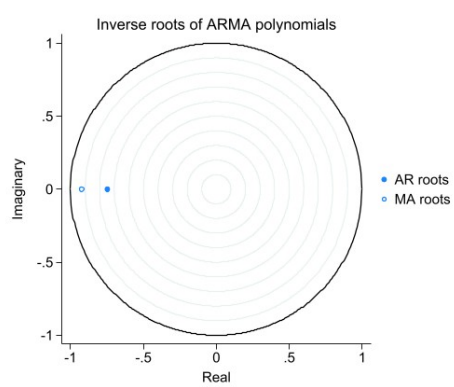

Fig. 7. Inverse roots of ARMA polynomials

VI. Forecasting

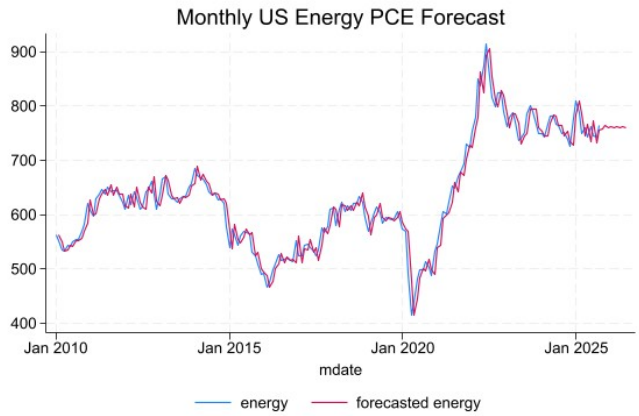

Fig. 8. Monthly US Energy PCE Forecast

TABLE XIII
Monthly US Energy PCE Forecast

Date	Forecasted Energy
2025m10	757.9545
2025m11	763.2967
2025m12	759.3027
2026m1	762.2887
2026m2	760.0563
2026m3	761.7253
2026m4	760.4775
2026m5	761.4104
2026m6	760.7130

Using the selected ARIMA(1,1,1) model, forecasts were generated for the next nine months, from October 2025 to June 2026. The forecasted values of monthly energy consumption are reported (Table XIII) and illustrated (Fig. 8), where the observed series and the predicted series are plotted together for comparison.

The forecasts suggest that energy consumption is expected to remain relatively stable over the forecast horizon, fluctuating around 760 billion dollars. No strong upward or downward trend is predicted in the short term, which is consistent with the dynamics captured by the differenced ARIMA model.

VII. Conclusion

In conclusion, the ARIMA(1,1,1) model provides a coherent representation of the data and yields reasonable short-term forecasts, making it an appropriate tool for analyzing and predicting monthly energy consumption.

VIII. References

U.S. Bureau of Economic Analysis, Personal consumption expenditures: Energy goods and services [DNRGRC1M027SBEA], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/DNRGRC1M027SBEA.